# Comprehensive Road Accident Analysis Report

## Executive Summary

- **Project Purpose:** This report delivers a detailed, data-driven evaluation of road traffic accidents to identify key factors influencing accident severity, infrastructure risks, environmental variables, and vehicle involvement.
- **Tool Used:** The dashboard and analysis were built entirely in Microsoft Excel using advanced formulas, pivot engine architectures, multi-layered data modeling, dynamic timeline scrubbers, and interactive slicers.
- **Dataset Scale:** The underlying dataset processes a total of 9,625 recorded casualties collected across multi-year traffic safety logs spanning from 2019 to 2022/2023.
- **Core Severity Finding:** The vast majority of incidents result in Slight Severity injuries (88.4%), while fatal and serious injuries combined account for over 1,114 victims, representing significant human loss and requiring targeted emergency medical planning.
- **Primary Infrastructure Hazard:** Single Carriageway roads represent the primary hotspot for collisions, accounting for more than 75% of all recorded casualties.
- **Environmental Impact:** Adverse weather conditions, particularly snow-covered and wet road surfaces, act as the major catalyst for high accident volumes during winter months.

## Detailed Breakdown of Key Metrics and KPI Scorecards

- **Total Casualties:** 9,625 total victims recorded across all analyzed incident events.
- **Fatal Severity Count:** 74 casualties, representing 1.84% of total incidents. This metric tracks critical collisions resulting in loss of life and serves as the highest priority for road safety interventions.
- **Serious Severity Count:** 1,040 casualties, representing 10.8% of total incidents. This covers severe injuries requiring hospitalization and intensive medical care.
- **Slight Severity Count:** 8,511 casualties, representing 88.4% of total incidents. This reflects high-frequency, lower-impact urban and suburban collisions resulting in minor injuries.
- **Casualties by Car:** 371 casualties, accounting for a 78.5% share within the specific passenger vehicle baseline cohort analyzed.

## Dataset Schema and Column Dictionary

- **Accident_Index:** A unique alphanumeric record key assigned to identify and track each distinct incident.

- **Accident_Date:** The specific date on which the collision occurred, enabling yearly, monthly, and seasonal aggregation.
- **Accident_Severity:** Categorical classification of the injury outcome, divided into Fatal, Serious, and Slight.
- **Number_of_Casualties:** The numeric count of individuals injured or killed in a single incident event.
- **Vehicle_Type:** The classification of the vehicle involved, such as Bus, Van, Motorcycle, Agricultural Tractor, or Car.
- **Road_Type:** The infrastructure layout where the collision took place, including Single Carriageway, Dual Carriageway, Roundabout, One Way Street, and Slip Road.
- **Road_Surface_Conditions:** The physical state of the road surface during the accident, categorized into Dry, Snow, or Wet.
- **Light_Conditions:** The ambient lighting level present at the time of collision, split into Daylight and Darkness.
- **Urban_or_Rural_Area:** The geographic environment of the crash location, categorized as Rural, Urban, or Unknown.
- **Weather_Conditions:** Meteorological observations during the incident, including Fine, Fog, Rain, Snow, and Other.
- **Wind_Conditions:** Atmospheric wind measurements, classified as High Winds, No High Winds, or Other.

# Detailed Visualization and Chart Breakdown

## 1. Key Performance Indicator Scorecards (Top Section)

- **Visual Structure:** Positioned prominently across the top row using custom donut visuals and large numerical callouts for immediate scanning.
- **Analytical Takeaway:** The visual highlights that 88.4% of casualties fall into the Slight category (8,511 victims), showing that modern vehicle safety features generally prevent high-speed mortality. However, the 1,114 combined fatal and serious casualties represent a heavy toll that requires focused policy intervention.

## 2. Casualties by Vehicle Type (Icon Graphic Matrix)

- **Vans:** Recorded 818 casualties, highlighting significant road exposure and risk among commercial delivery and transport vehicles.
- **Motorcycles:** Recorded 772 casualties, showing high vulnerability and risk among two-wheeler riders due to minimal physical protection.
- **Buses:** Recorded 371 casualties, reflecting a high victim density per single collision event due to passenger capacity.
- **Agricultural Tractors:** Recorded 31 casualties, representing localized risks on rural roads involving heavy farm machinery.

- **Cars:** Recorded 73 casualties within the specific benchmark passenger subset analyzed in this section.

## 3. Casualties Monthly Trends (Multi-Year Line Chart)

- **Visual Structure:** A comparative line chart mapping seasonal fluctuation month-by-month across 2019, 2020, 2021, and 2022.
- **Winter Spikes:** Consistent peaks occur in early winter (January and February) and late winter (November and December), where monthly casualties frequently exceed 1,200 to 1,400 during high-volume years like 2021 and 2022.
- **Mid-Year Declines:** A noticeable drop in casualty counts occurs from April through September, coinciding with warmer weather, longer daylight hours, and dry road surfaces.

## 4. Casualties by Road Type (Horizontal Bar Chart)

- **Single Carriageway:** 7,254 casualties (75.37% of total). This is by far the most dangerous road layout due to undivided opposing traffic lanes and higher risks of head-on collisions.
- **Dual Carriageway:** 1,684 casualties (17.50% of total). Shows a significantly lower casualty rate despite higher travel speeds, proving the safety benefit of central median dividers.
- **Roundabout:** 428 casualties (4.45% of total). Represents low-speed junction collisions caused by failing to yield.
- **One Way Street:** 111 casualties (1.15% of total). Shows minimal accident occurrence due to simplified traffic flow.
- **Slip Road:** 98 casualties (1.02% of total). Occurs primarily during high-speed highway merging maneuvers.
- **Unknown Road Type:** 50 casualties (0.52% of total).

## 5. Casualties by Road Surface Conditions (Vertical Bar Chart)

- **Snow Surface:** 7,741 casualties. Represents the largest share of accidents, proving that freezing weather, black ice, and loss of tire traction are the single biggest catalyst for crashes.
- **Wet Surface:** 1,809 casualties. Caused by hydroplaning, reduced tire grip, and increased braking distances during rainfall.
- **Dry Surface:** 75 casualties. Represents a tiny fraction of total casualties in this dataset, emphasizing that adverse weather is the primary driver of crashes here.

## 6. Area Wise Casualties (Comparative Column Chart)

- **Rural Areas:** 5,112 casualties (53.11% of total). Generates higher overall casualties due to higher speed limits, narrower lanes, lack of street lighting, and delayed emergency response times.
- **Urban Areas:** 4,512 casualties (46.88% of total). Accounts for high-frequency collisions occurring at lower speeds due to heavy traffic congestion, pedestrian activity, and frequent intersections.
- **Unknown Area:** 1 casualty (0.01% of total).

### 7. Casualties by Road Lights Available (Pie Chart)

- **Daylight Conditions:** 5,300 casualties (55.07% of total). Higher total volume due to heavily elevated daytime traffic density.
- **Darkness Conditions:** 4,325 casualties (44.93% of total). Represents a disproportionately high risk level given that nighttime traffic volumes are significantly lower, driven by reduced visibility, driver fatigue, and higher speeds.

## Interactive Dashboard Features and Control Systems

- **Weather Condition Slicer:** Allows users to dynamically filter the entire dashboard by Fine, Fog, Rain, Snow, or Other weather states.
- **Wind Condition Slicer:** Enables instant filtering between High Winds, No High Winds, and Other atmospheric conditions.
- **Timeline and Date Scrubber:** Gives users the ability to slice data across specific years (2021, 2022, 2023) or custom date ranges to track progress over time.

## Comprehensive Statistical Conclusions and Policy Recommendations

### Core Conclusions

- **Infrastructure Vulnerability:** Single Carriageway roads account for 3 out of every 4 casualties (7,254 victims), making them the primary physical infrastructure hazard.
- **Weather Sensitivity:** Over 80% of casualties occur on snow-covered roads, demonstrating that severe winter weather dramatically increases collision frequency.
- **Rural Road Risk:** Rural areas generate more casualties (5,112) than urban environments (4,512), proving that higher speeds on undivided roads outweigh urban traffic density risks.

### Strategic Recommendations

- **Infrastructure Modifications:** Prioritize installing center rumble strips, median barriers, and additional overtaking lanes on high-volume Single Carriageway roads to eliminate head-on collisions.
- **Winter Road Management:** Increase municipal budget allocations for road salting, gritting, and snow plowing along major rural corridors ahead of peak winter months (November through February).
- **Speed Enforcement in Rural Zones:** Implement automated speed cameras and improved retro-reflective road signage along high-risk rural routes.
- **Targeted Safety Campaigns:** Launch targeted safety education programs for commercial van drivers (818 casualties) and motorcycle riders (772 casualties), emphasizing defensive driving techniques in low-traction environments.